6. The Gamma and Zeta functions

Def I.6.1. For real $s>0$, the Gamma function $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1}\, dt$ (*)

Rmk. Since $\forall s>0$ $\int_0^\infty e^{-t}t^{s-1}\,dt$ converges, $\Gamma(s)$ is well-defined

Prop. I.6.2. $\Gamma(s)$ can extend to an analytic function in $\mathrm{Re}(s)>0$, and be still given by (*)

Pf: We show that the integral defines an analytic function on $S_{\delta,M} = \{\delta < \mathrm{Re}(s) < M\}$

Since $f(t,s) = e^{-t}t^{s-1}$ • is holomorphic for s when fixing t
• is continuous on $[0,\infty)\times\{\mathrm{Re}(s)>0\}$

Lemma I.6.3. (1) $\{f_n\} \subseteq H(D)$, $f_n \to f$ uniformly in every compact set of D.
Then $f \in H(D)$

(2) If $F(z,s)$, $(z,s) \in \Omega \times [a,b]$ satisfying ① $F(z,s) \in H(\Omega)$ $\forall s \in [a,b]$
② $F(z,s) \in C(\Omega \times [a,b])$

Then $\int_a^b F(z,s)ds$ is holomorphic in $H(\Omega)$

Pf: (1) $f_n \to f \Rightarrow f$ is continuous on $K \subseteq D$ compact. $\int_T f_n \to \int_T f \Rightarrow \int_T f(z)\,dz = 0$
By Morera's Thm, we have $f \in H(K)$ Thus $f \in H(D)$

(2) $F_n(z,s) = \frac{b-a}{n} \sum_{k=0}^{n} F(z, \frac{na+k(b-a)}{n})$ Then $\{f_n\} \to f$ uniformly $f_n \in H(\Omega)$
$\Rightarrow f \in H(\Omega)$

Back to the proof: $\forall \varepsilon > 0$ $\int_\varepsilon^\xi e^{-t}t^{s-1}\,dt \in H(\mathrm{Re}(s)>0)$.

Now, we choose $\varepsilon = \frac{1}{n}$ $F_n(z) = \int_{\frac{1}{n}}^{n} e^{-t}t^{s-1}\,dt$.

We show that $\{F_n(s)\} \to \Gamma(s)$ uniformly on $\delta < \mathrm{Re}(s) < M$

$$|\Gamma(s) - F_n(s)| = \left|\int_0^\infty e^{-t}t^{s-1}\,dt - \int_{\frac{1}{n}}^{n} e^{-t}t^{s-1}dt\right| \le \int_0^{\frac{1}{n}} |e^{-t}t^{s-1}|\,dt + \int_n^\infty |e^{-t}t^{s-1}|\,dt \quad (**)$$

Let $\mathrm{Re}(s) = 6$. Then $(**) = \int_0^{\frac{1}{n}} e^{-t}t^{6-1}\,dt + \int_n^\infty e^{-t}t^{6-1}dt < \left(\frac{1}{n}\right)^6 \cdot \frac{1}{6} + C\int_n^\infty e^{-\frac{t}{2}}dt$

where $\delta < 6 < M \Rightarrow F_n(s) \to \Gamma(s)$ uniformly for $\delta < \mathrm{Re}(s) < M$.

$\Rightarrow \Gamma(s) \in H(\delta < \mathrm{Re}(s) < M) \Rightarrow \Gamma(s) \in H(\mathrm{Re}(s)>0)$. □

Prop I.6.4. If $\mathrm{Re}(s)>0$, then $\Gamma(s+1) = s\Gamma(s)$

Proof: $\int_\varepsilon^{\frac{1}{\varepsilon}} \frac{d}{dt}(e^{-t}t^s)\,dt = -\int_\varepsilon^{\frac{1}{\varepsilon}} e^{-t}t^s\,dt + s\int_\varepsilon^{\frac{1}{\varepsilon}} e^{-t}t^{s-1}\,dt$

Let $\varepsilon \to 0^+$, we have $0 = -\Gamma(s+1) + s\Gamma(s)$ □. $\left(\int_\varepsilon^{\frac{1}{\varepsilon}} f'(t)\,dt = f(t)\Big|_\varepsilon^{\frac{1}{\varepsilon}} \to 0 \ (\varepsilon \to 0^+)\right)$

Thm I.6.5. $\Gamma(s)$ initially defined on $\mathrm{Re}(s)>0$ has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at $s=0,-1,-2,\dots$ with the residue at $s=-n$ $\frac{(-1)^n}{n!}$

proof: For $\mathrm{Re}(s)>-1$, define $F_1(s)=\frac{\Gamma(s+1)}{s}$

Since $\Gamma(s+1)\in H(\mathrm{Re}(s)>-1)$, $\Gamma(1)=\int_0^\infty e^{-t}dt=1 \Rightarrow F_1(s)$ has a simple pole at $s=0$

Similarly, for $\mathrm{Re}(s)>-m$ define $F_m(s)=\frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}$

has simple poles at $s=0,-1,\dots,-m+1$

$\mathrm{Res}_{s=-n} F_m(s)=\frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)}=\frac{(-1)^n}{n!}$ $\square$

Rmk. From the above theorem, $\Gamma(s)=\int_0^1 e^{-t}t^{s-1}dt+\int_1^\infty e^{-t}t^{s-1}dt$ $\forall\ \mathrm{Re}(s)>0$

$\int_1^\infty e^{-t}t^{s-1}dt$ can define an entire function.

$=\int_0^1\left(\sum_{n=0}^\infty \frac{(-t)^n}{n!}t^{s-1}\right)dt+\int_1^\infty e^{-t}t^{s-1}dt$.

$\overset{\text{uniform}}{=}\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)}+\int_1^\infty e^{-t}t^{s-1}dt$ (*)

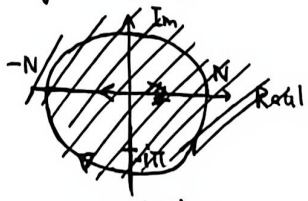
Hence (*) defines a meromorphic function on $\mathbb{C}$ with residue at $s=-n$ $(n\geq 0)$: $\frac{(-1)^n}{n!}$

Thm I.6.6. $\forall s\in\mathbb{C}$. $\Gamma(s)\Gamma(1-s)=\frac{\pi}{\sin\pi s}$

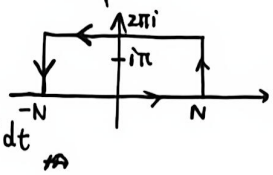
Pf: Since $\Gamma(s)\Gamma(1-s)$ is meromorphic on $\mathbb{C}$ with simple poles at all integers. same as $\frac{\pi}{\sin\pi s}$

We just need to show $\forall\ 0<s<1$, the identity holds. Now $\forall\ 0<a<1$.

$\int_0^\infty \frac{v^{a-1}}{1+v}dv=\int_{-\infty}^\infty \frac{e^{ax}}{1+e^x}dx=\lim_{N\to\infty}\int_{-N}^N \frac{e^{ax}}{1+e^x}dx$



$\int_{-N}^N \frac{e^{ax}}{1+e^x}dx=\int_{\Gamma_N} f+\left(\int_{-N}^N \frac{e^{ax}}{1+e^x}dx\right)e^{2\pi i a}-\int_0^{2\pi}\frac{e^{a(N+it)}}{1+e^{N+it}}dt+\int_0^{2\pi}\frac{e^{a(-N+it)}}{1+e^{-N+it}}dt$



$\Rightarrow \int_0^\infty \frac{v^{a-1}}{1+v}dv=\frac{\pi}{\sin\pi a}$

$\Rightarrow \Gamma(s)\Gamma(1-s)=\int_0^\infty e^{-t}t^{s-1}\left(t\int_0^\infty e^{-vt}(vt)^{-s}dv\right)dt=\int_0^\infty\int_0^\infty e^{-t[1+v]}v^{-s}dv\,dt$

$=\int_0^\infty \frac{v^{-s}}{1+v}dv=\frac{\pi}{\sin\pi s}$ $\square$.

Coro I.6.7. $\Gamma(\frac{1}{2})=\sqrt{\pi}$

^ u+iv 为复数，h 定义在 $\mathbb{R}$ 上的可导实函数且 $u=hov$. 证明：f 为幂数

Def. I.6.8. (1) For real $s>1$ The zeta function: $\zeta(s)=\sum_{n\geq 1}\frac{1}{n^s}$ (Immediately we have $Re(s)>1$)

(2) For real $t>0$ The theta function: $\theta(t)=\sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$

(3) For ~~real~~ $Re(s)>1$. The xi function: $\xi(s)=\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\zeta(s)$.

In Analytic Number Thoory Coarse,

We proved the following equation:

Thm I.6.9. For $Re(s)>1$, $\xi(s)=\xi(1-s)=\int_0^\infty u^{\frac{s}{2}-1}(\theta(u)-1)\,du$.

Thus, we have $\zeta(s)=\frac{\pi^{\frac{s}{2}}\xi(s)}{\Gamma(\frac{s}{2})}$. where $\frac{1}{\Gamma(\frac{s}{2})}$ is entive with simple zeros at $s=0,-2,-4,\dots$

$\xi(s)=\frac{1}{s-1}-\frac{1}{s}+\int_1^\infty \underbrace{\left(u^{-\frac{s}{2}-\frac{1}{2}}+u^{\frac{s}{2}-1}\right)}\frac{\theta(u)-1}{2}\,du$. with simple poles at $s=0$, $s=-1$

↘ entire

$\Rightarrow \xi(s)$ can be extended to a meromorphic function on $\mathbb{C}$ with a simple pole at $s=-1$ with the residue 1.